Closed-Loop Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Energy Absorbers

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Tensegrity-inspired architectures, in which rigid struts are suspended within a continuous flexible network, can exhibit tunable nonlinear mechanical responses and high strength-to-weight ratios, motivating their use in lightweight protective and energy-absorbing structures. Multi-material fused deposition modeling (FDM) can co-print rigid (PLA) struts and flexible (TPU) elements in a single build, but the resulting design space—spanning strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling—is too large to explore efficiently by trial-and-error experimentation.

This work presents a closed-loop experimental campaign that uses Bayesian optimization (BO) to develop multi-material 3D-printed tensegrity-inspired structures for improved energy absorption under quasi-static compression and drop-weight impact loading. The key contribution is a design–print–test workflow in which experimental measurements are used directly to update a Gaussian-process surrogate model, enabling sequential design selection without relying on finite-element simulation. The surrogate is trained on peak transmitted force, specific energy absorption (SEA), and compaction efficiency, and a batch acquisition function recommends the next set of candidate designs after each test round. The workflow combines automated model updating and experiment selection with manual specimen handling and test setup.

We define a parameterized family of unit cells using a core-wrapping architecture in which rigid PLA struts are enclosed by continuous TPU skins to improve interfacial interlocking. Design variables include strut diameter and length, tension-element width and thickness, strut count per unit cell, and tiling pattern. Specimens are fabricated on a multi-material FDM system and tested under quasi-static compression and instrumented drop-weight impact.

We will report how BO-selected designs evolve across iterations, how the surrogate model changes as data accumulate, and what Pareto-efficient trade-offs emerge between SEA and peak transmitted force within the explored design space. We will also discuss practical lessons for operating the closed-loop workflow, including print-failure handling, batch-to-batch variability, and the exploration–exploitation trade-off.